Aditi Simha

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Summary

Software Engineer with 4+ years designing and delivering resilient, cloud-native microservices, real-time data pipelines, and production-grade REST APIs across regulated enterprise environments. Owns the full backend lifecycle of architecture, API design, deployment, and performance tuning, with a proven track record modernizing high-revenue legacy systems.

Skills

- **Languages:** Java, Python, C++, SQL (PostgreSQL, Oracle DB, MySQL), JavaScript
- **Frameworks & Tools:** Spring Boot, Django, FastAPI, RESTful Web Services, Swagger/OpenAPI, Hibernate
- **Cloud & DevOps:** Docker, Kubernetes, Jenkins, Spinnaker, AWS (EC2, S3, Lambda), GCP (Pub/Sub), CI/CD, Maven
- **Data & Architecture:** Microservices, Distributed Systems, Kafka, Redis, Elasticsearch, LangChain, Vector Search
- **Testing & Tools:** JUnit (TDD), JBehave (BDD), Mockito, Splunk, New Relic, Git (Gitflow), SVN, Jira

Work Experience

Software Engineer, ***Live Music Project*** — **Oct 2025 - Present**

- Replaced a fully manual weekly/monthly email workflow with an automated self-service digest system using **Django PostgreSQL** with **GeoDjango** (**PostGIS**) spatial queries to match content to regional subscribers.
- Automated a manual nightly email dispatch into a fault-tolerant background pipeline with **Redis-backed Huey** workers for scheduled, asynchronous processing.
- Enabled scalable transactional email delivery with zero disruption to existing production infrastructure, integrated via **MailChimp SMTP** within a containerized **Docker** environment using isolated mail-connection handling.

Software Engineer II, ***Oracle Cerner*** — **May 2022 - Aug 2023**

- Migrated RadNet's legacy C++ image retrieval engines to **Spring Boot microservices** for Oracle Cerner's **SaaS modernization** initiative, reducing clinical image retrieval latency by 60% across 12 IDN networks (200+ facilities).
- Designed a greenfield diagnostic reporting layer **REST API** from scratch using **Swagger**, **Java**, and **OAuth 2.0**, defining integration contracts and data models used by 6+ consuming services in real-time clinical workflows.
- Optimized high-volume **Oracle** and **MySQL** data-access workflows, accelerating API response times by 40% eliminating query bottlenecks on high-volume production traffic.
- Built automated **CI/CD** and validation pipelines using **Jenkins**, **JUnit**, and **JBehave**, increasing release deployment reliability by 25% and cutting regression cycle time for critical healthcare systems.
- Drove distributed tracing strategy using **Splunk** and **New Relic**, establishing alerting baselines that caught production anomalies early and significantly reduced system MTTR.

Software Engineer I, ***Oracle Cerner*** — **Jul 2020 - May 2022**

- Boosted backend throughput by 20% for high-revenue radiology systems by refactoring performance-critical **C++** modules, directly improving SLA compliance across enterprise clients.
- Led migration from **SVN** to **Git**, standardizing Gitflow branching across the Radiology codebase and reducing merge conflicts by 30%, enabling parallel feature development across distributed teams.
- Standardized service integration contracts and API documentation across three engineering squads, reducing cross-team integration errors by 40% and accelerating delivery timelines.
- Owned end-to-end software lifecycle: **BDD** test authoring, defect triage, and production deployment readiness, ensuring zero-defect releases for regulated healthcare environments.

Software Intern, ***Oracle Cerner*** — **Jan 2020 - Jun 2020**

- Built **Node.js** web crawlers and internal metadata indexing tools backed by **Elasticsearch**, improving artifact searchability and discoverability for 500+ engineers across the organization.

Projects

Pipeline Runner: GPU-Scheduled AI Pipeline Service (FastAPI · asyncio · Pydantic)

- Built a type-safe API validating pipeline chains at submit-time to block invalid jobs before GPU consumption, with cooperative cancellation and per-step failure attribution.
- Built a deterministic load-testing framework (seeded RNG + virtual clock) simulating hours of bursty GPU traffic in under a second, with live cost, utilization, and p50/p95 latency metrics.

Real-Time Trimet Pipeline (Python · PostgreSQL)

- Built event-processing workflows handling 1.5M+ real-time GPS events using Pub/Sub and Kafka ingestion pipelines with optimized batching strategies and fault-tolerant consumer groups.
- Designed PostgreSQL schema and partitioning strategy to support sub-second analytical queries over continuously ingested geospatial data at scale.

Education

Master of Science, Computer Science — 2025
Portland State University — GPA: 3.85

Bachelor of Engineering, Computer Science — 2020
Visvesvaraya Technological University — GPA: 4.0